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the European Union

MED&BS RDBFIS-III

ESTABLISHMENT OF THE REGIONAL DATABASE FOR THE MEDITERRANEAN AND BLACK SEAS

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23/9/2025



RDBFIS: Overview of RDBFIS Phases I, II, III

Phase I

Start: 01/01/2021

End: 28/02/2023

MARE/2020/08 - SI2.839444

DEVELOPMENT OF THE REGIONAL DATABASE FOR THE
MEDITERRANEAN & BLACK SEAS (Med&BS RDBFIS)

This project has financed under the European Maritime and Fisheries Fund (EMFF)



Phase II

Start: 01/04/2023

End: 31/03/2025

CINEA/EMFAF/2021/3.1.2/03/SC04/SI2.881222

HOSTING, MAINTENANCE AND FURTHER DEVELOPMENT OF THE REGIONAL
DATABASE FOR THE MEDITERRANEAN AND BLACK SEAS



Non partners but with a significant contribution



Phase III

Start: 01/04/2025

End: 21/05/2026

CINEA/EMFAF/2024/3.1/01/SC08

ESTABLISHMENT OF THE REGIONAL DATABASE FOR THE MEDITERRANEAN
AND BLACK SEAS



Partners

Hellenic Centre for Marine Research (HCMR) – Project Coordinator
Stefanos Kavadas – Scientific IT Coordinator
University of Athens (UOA)
Coispa Tecnologia & Ricerca Scarl (COISPA)
Consorzio per il Centro Interuniversitario di Biologia Marina ed Ecologia Applicata G. Bacci (CIBM)
Italian National Research Council (CNR) (Istituto per le Risorse Biologiche e le Biotecnologie Marine (CNR-IRBIM) and Istituto per il Rilevamento Elettromagnetico dell'Ambiente (CNR-IREA))
Nisea Società cooperative (NISEA)
University of Athens (UOA)
University of Thessaly (UTH)
Hellenic Agricultural Organization Demeter (HAO Demete)
Ministry of Agriculture, Rural Development and Environment of Cyprus – Department of Fisheries and Marine Research, Republic of Cyprus (DFMR)
National Institute for Marine Research and Development “Grigore Antipa” (INCDM)
Institute of Oceanography and Fisheries (IOF)
Institute of Oceanology BAS (IO-BAS)
MEDIAS expert (Tarek Hattab, Ifremer)
CSIC/IEO

Non-partner with a significant contribution

Executive Agency for Fisheries and Aquaculture, Bulgaria (Simona Vasileva NICHEVA)
Ministry of Agriculture, Directorate of Fisheries, Croatia (Ivana VUKOV)
Fisheries Research Unit, Department of Fisheries and Aquaculture, Malta (Jurgen Mifsud)
Maurizio GIBIN, Maciej ADAMOWICZ, Maksims KOVSARS quality (checks on spatial information and mapping for the FDI EWG)

Aggregated data

- Med&BS
- FDI (Tables A, B, G, H, I, J)
- AER
- GFCM/DCRF

MEDITS (TA, TB, TC)

Detailed biological & landings data

MEDIAS



A straight-forward process to upload data

Step 1. Syntax check (validation scheme)

Step 2. Consistency check (quality control routines)

Step 3. Upload (update RDBFIS)



Med&BS RDB

- Med&BS dc
- FDI dc
- GFCM/DCRF dc
- Detailed bio data
- MEDITS, MEDIAS, Eggs&Larvae
- EU Fleet
- Stomach content
- PETS
- Alien species
- Recreational
- Coding
- Environmental
- Spatial data

Integrated Components

- ✓ UI
- ✓ Validation schemes
- ✓ RDBqc
- ✓ RDBprocessing
- ✓ RoME, RoMEBS
- ✓ BioIndex
- ✓ EU fleet analysis
- ✓ MCDA for SSF
- ✓ FDI spatial checks
- ✓ Mapping
- ✓ Stomach content & Eggs&Larvae data entry forms

Aim

- ❖ Continue the **hosting, maintenance, fine-tuning and further development** of RDBFIS;
- ❖ Finalize the **population of RDBFIS with data** and agree on **data call for the future**;
- ❖ Work towards using RDBFIS **for submission to the main data calls and reporting obligations**;
- ❖ Provide **support to the users** and ensure the migration to the **permanent host** once a decision is taken.

Task 0 – Project Management and Coordination

Task 1 Provision of a temporary solution for hosting and maintenance of RDBFIS

Task 2 Population of RDBFIS with MS data and future data call

Task 3 Submission of data to end users through RDBFIS

Task 4 Fine-tuning and further development

Task 5 Provision of support to the users

Task 6 Advise on way forward, in terms of improving the RDB for the region

Task 7 Participation in meetings, where the work of this contract needs to be disseminated



Task 8: Prepare a hand over file to facilitate the transition of RDBFIS to DG MARE, RCG Med&BS and the final host

Project and calendar month													
2025					2026								
01 Apr.	02 May	03 June	04 July	05 Aug.	06 Sep.	07 Oct.	08 Nov.	09 Dec.	10 Jan.	11 Feb.	12 Mar	13 Apr.	14 May

Actions already performed

❖ Participation in **meetings**:

- MEDIAS SC
- MEDITS SC
- RCG ECON
- RCG LP
- SC RDB
- RCG MED&BS

❖ **Dedicated workshops/meetings** relevant to fine tuning & further development are organized and actions have started.

TASK 4: Fine-tuning and further development of RDBFIS

1-Regional database maintenance and restructuring

S. Kavadas, I. Maina (HCMR)
M: 1-13

2- Update/revision of existing data quality tools

I. Bitetto, W. Zupa (COISPA)
M: 2 - 5

3-Development of data quality tools: MEDIAS

S. Kavadas, A. Kapelonis (HCMR), T. Hattab
M: 2- 8

4-Further development & adjustments of tools4MCDA

I. Maina (HCMR), S. Kavadas
M: 8-1

5-Improvement & further development of the spatial visualization (mapping) component

I. Dokos , I. Maina (HCMR)
M:1 to 9)

6- Improvement and further development of the existing validation scheme and consistency checks

S. Kavadas, I. Maina (HCMR), I. Chamodrakas(UOA)
1 to 12)

7-Fleet analysis (yearly updates), improvement of spatial checks for FDI

S. Kavadas, I. Chamodrakas (UOA)
M: 5 to 6

8-Integration of sampling optimization in RDBFIS

I. Bitetto W. Zupa (COISPA)
M: 1 to 12

9-Common raising procedures for the Med&BS biological data

M. Pantazi (HCMR), K. Touloumis (FRI)
M: 1 to 4

10-stat4RDBFIS software Package development

V. Sgardeli (HCMR), A. Mannini (CNR)
M: 4 to 12

11-Development of the MAPAFISH-MED tool – Fishing activities associated with Marine Protected Areas (MPAs)

V. Sgardeli, D. Damalas (HCMR)
M: 2, 3, 12, 13

12-Fine-tuning

I. Chamodrakas (UoA), S. Kavadas, K. Kalkavouras (HCMR)
M: 1 to 13

Activity 2: Update/revision of existing data quality tools

Coordinators: Isabella Bitetto (COISPA), Walter Zupa (COISPA)

- ✓ further improving the functionalities of the **RoME** and **RDBqc** quality check tools, **BioIndex** data analysis tool where needed.
- ✓ address issues identified during earlier projects and training sessions by leveraging user feedback and insights gained from practical applications of these tools.
- ✓ enhancing existing features and introducing new functionalities to meet evolving requirements.

The screenshot displays the Med&BS RDBFIS web application. On the left is a dark sidebar menu with options: Data Calls, Surveys (highlighted in red), MEDITS, MEDIAS, Stomach Contents, Eggs & Larvae, Alien/Invasive Species, Recreational Fisheries, Litter, EU Fleet Data, Spatial Analysis, Environmental, Codification System, and User Settings. The main content area is titled 'MEDITS' and has tabs for 'Tabular Visualization', 'Import CSV data', and 'BioIndex analysis' (which is active). Below the tabs is a form for 'BioIndex analysis' with various input fields: Country (dropdown), Area (dropdown), Species (dropdown), Start Year (2020), End Year (2020), Depth Lower (10), Depth Upper (800), Depth Lines (a list with 50, 100, 200, 500, and 1000, where 200 is selected), Latitude lower limit (34), Latitude upper limit (45.7), Longitude lower limit (-5.5), Longitude upper limit (35), Rec threshold (200), Spaw threshold (210), and Haul threshold (30). A blue 'Run BioIndex' button is at the bottom right of the form. At the bottom of the page, there is a footer with the text 'RDBFIS: Financed by EMFF (2021-2022) and CINEA/EMFAF (2023-2025)' and the European Union flag.

- ❖ started to improve functionality – incorporate MEDITs Handbook updates

Activity 3: Development of data quality tools: MEDIAS

Coordinators: Stefanos Kavadas, Aris Kapelonis-HCMR, Tarek Hattab

- ✓ During the implementation of RDBFIS II, and based on feedback from MEDIAS experts, a robust **validation scheme** and comprehensive **data quality checks** have been designed for **acoustics processing data, trawl pelagic samples, and CTD casts**.
- ✓ In this project we will focus on refining the code and develop an open-source **R package** to be integrated into the **RDBFIS** system.

The screenshot displays the Med&BS RDBFIS web application. The sidebar on the left contains a menu with items: Data Calls, Surveys (highlighted), MEDITS, MEDIAS, Stomach Contents, Eggs & Larvae, Alien/Invasive Species, Recreational Fisheries, Litter, EU Fleet Data, Spatial Analysis, Environmental, Codification System, and User Settings. The main panel is titled 'MEDIAS' and includes tabs for 'Tabular Visualization', 'Import Acoustics CSV data', 'Import Pelagic Trawl CSV data', and 'Import CTD CSV data'. Under 'Tabular Visualization', there is a 'Select a table' dropdown menu currently set to 'MEDIAS acoustics survey identity', with an 'Open Advanced Query Form' button next to it. Below this is a 'Data Table' with the following columns: Survey Name, Cd Country, Gsa, Survey Area, Cd Centre, Cd Inst, Cd Vessel, and Sampling Year. Each column has a 'Filter by' input field. The table contains five data rows:

Survey Name	Cd Country	Gsa	Survey Area	Cd Centre	Cd Inst	Cd Vessel	Sampling Year
MEDIAS	GRC	GSA20	ElonSea-GRC-IMBRIW	HCMR	IMBRIW	PHI	2021
MEDIAS	GRC	GSA20	ElonSea-GRC-IMBRIW	HCMR	IMBRIW	PHI	2022
MEDIAS	GRC	GSA22	AegSea-GRC-IMBRIW	HCMR	IMBRIW	PHI	2021
MEDIAS	GRC	GSA22	AegSea-GRC-IMBRIW	HCMR	IMBRIW	PHI	2022
MEDIAS	FRA	GSA7	GLion-FRA-IFREMER	IFREMER	IFREMER	EUR	2023

At the bottom of the table, it indicates '5 total' rows. A green 'Export to CSV' button is located below the table. The footer of the page states: 'RDBFIS: Financed by EMFF (2021-2022) and CINEA/EMFAF (2023-2025)' with the European Union flag logo.

- ❖ **MEDIAS R package:** dedicated workshop was performed on **16/5/2025** and implementation of the software and beta version of functions has started.

Med&BS RDBFIS-III: Fine-tuning and further development of RDBFIS (TASK 4)

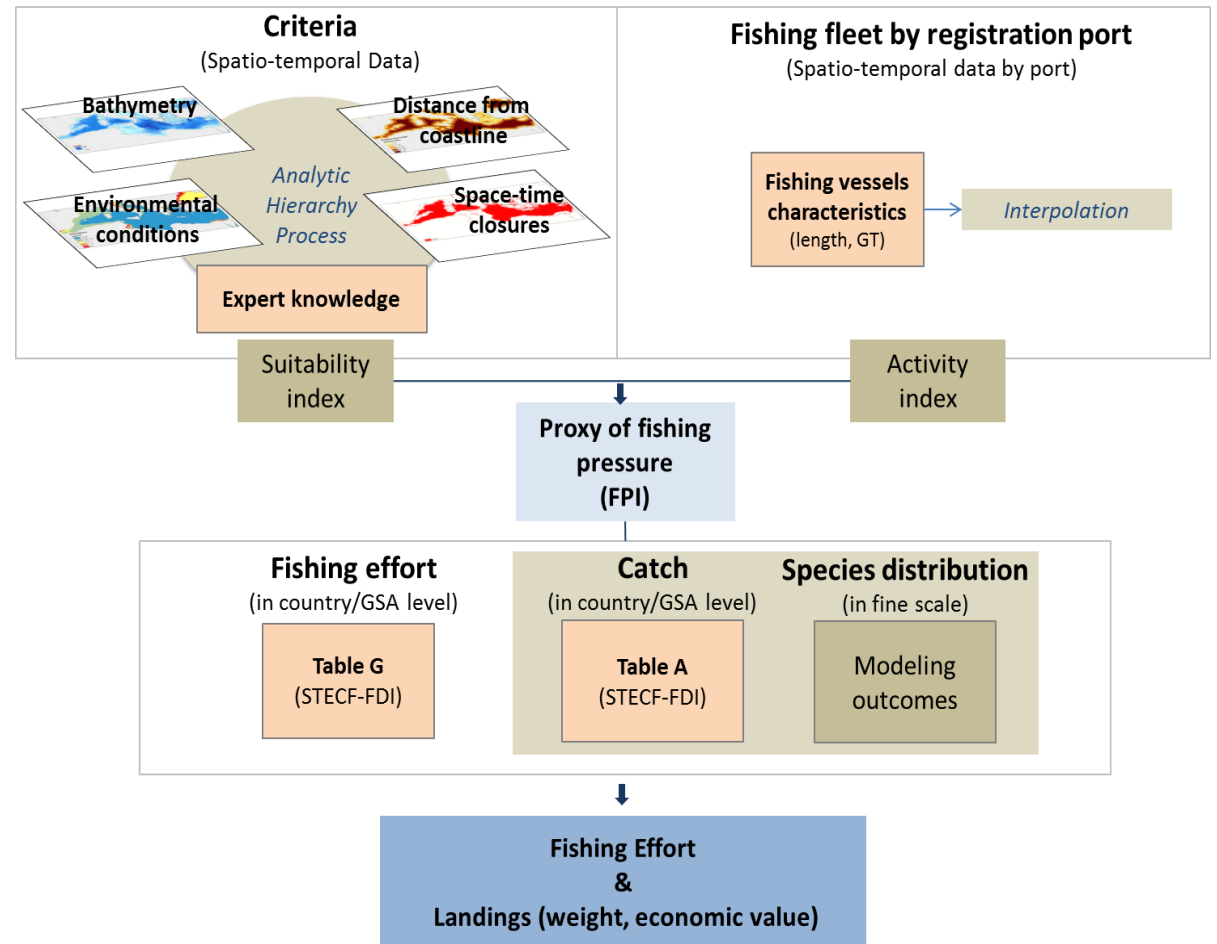
Activity 4: Further development & adjustments of the spatial estimation of fishing effort and landings from Small Scale Fisheries: tools4MCDA

Coordinators: Irida Maina (HCMR), Stefanos Kavadas

tools4MCDA : estimating the spatial distribution of fishing effort in **small-scale fisheries** (vessels <12 m) often lack monitoring equipment and is **integrated to RDBFIS**.

New improvements are related to the:

- ✓ Enhancement/optimization of scripts
- ✓ Expand user options
- ✓ Improve the visualization
- ✓ Provide analytical instructions



The screenshot shows the Med&BS RDBFIS web interface. The left sidebar contains a menu with options like Data Calls, FDI, GFCM/DCRF, Med & BS, RCG, AER Socioeconomic, GFCM Socioeconomic, STECF/GFCM, Surveys, and Biomass Contents. The main content area is titled 'FDI' and includes tabs for Tabular Visualization, Import CSV data, Processing, and MCDA. The MCDA tab is active, showing a form for selecting FPI per quarter and gear CSV file, and FPI species distribution CSV file. The form includes fields for Country (Greece), Start Year (2020), and End Year (2020). There are buttons for 'Syntax & Consistency Checks' and 'Restart Procedure'. Below the form, the 'MCDA Analysis Procedure' section shows 'Stage Results'.

Activity 6: Improvement and further development of the existing validation scheme and consistency checks (MEDIAS, socio economic, cross validations between the EC datacalls)

Coordinators: Stefanos Kavadas, Irida Maina (HCMR), Ioannis Chamodrakas(UOA)

- ✓ This activity focuses on the improvements and further development of the existing **validation scheme and consistency checks** established within the RDBFIS.
- ✓ The enhancements primarily target key data sets such as **MEDIAS, socio-economic datasets, and cross-validation processes** between various EC data calls.
- ✓ By refining these mechanisms, the project aims to strengthen **data quality, reliability, and interoperability** across multiple data domains.
- ✓ These advancements will contribute to ensuring **robust, harmonized, and actionable** data sets.

Activity 7: Fleet analysis (yearly updates), improvement of spatial checks for FDI (objective 6)

Coordinators: Stefanos Kavadas, Ioannis Chamodrakas (UOA)

✓ **Fleet analysis tool** will be updated **annually** using the latest data from the **EU fleet register portal**. Spatial identification of ports and the analysis of data at different geographical levels (e.g., NUTS2, NUTS3, registration fishing port).

✓ **FDI spatial checks**, the improvements made by developers to quality checks on spatial effort and landings data (submitted for the FDI data call), **updates** to related shapefiles (e.g., coastline, geographical divisions), **will be incorporated** into the RDBFIS.

The screenshot displays the 'Med&BS RDBFIS' web application interface. On the left is a dark sidebar with a menu including 'Data Calls', 'Surveys', 'Stomach Contents', 'PETS', 'Alien/Invasive Species', 'Recreational Fisheries', 'Litter', 'EU Fleet Data' (highlighted in red), 'Fleet Vessels', 'Spatial Analysis', 'Environmental', 'Codification System', and 'User Settings'. The main content area is titled 'VESSELS' and contains a 'Fleet Analysis' section. This section has two tabs: 'Fleet Analysis' (active) and 'Fleet Analysis Results'. The 'Fleet Analysis' tab is divided into two steps: '1 Fleet Analysis - Query Parameters' and '2 Fleet Analysis Results'. Under step 1, there are several configuration options: 'Query on' set to 'Active Fleet', 'Vessel Length Segmentation' with radio buttons for 'Med&BS' (selected), 'GFCM', 'Baltic Sea', and 'Other waters', 'Type of Query' with a dropdown for 'Year' set to '2023', and a 'Between' date range from '1991' to '2023'. There are also filters for 'vessel characteristics' and 'Results Geographically', both set to 'No', and a 'Group Results' option set to 'No'. An 'Execute Fleet Analysis' button is at the bottom of the form. The footer of the page includes the text 'RDBFIS: Financed by EMFF (2021-2022) and CINEA/EMFAF (2023-2025)' and the European Union flag.

Activity 8: Integration of sampling optimization in RDBFIS

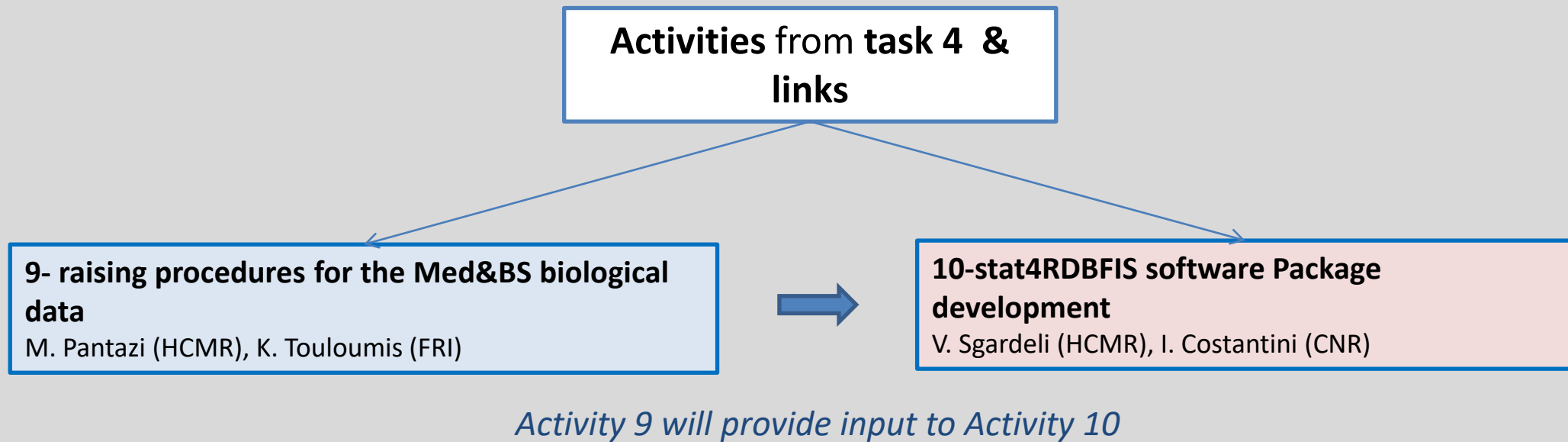
Coordinators: Isabella Bitetto (COISPA), Walter Zupa (COISPA)

A new specific component will be included in RDBFIS to allow the application of **the R tools (SDTool and BioSim tool) for biological sampling optimization**, developed in MARE/2014/19 Med&BS and STREAM projects.

The integration in RDBFIS will allow to use the detailed data in **RCG Med & BS format** stored in RDBFIS including:

- ✓ **Creation of the sampling optimization R package** and storage on the RDBFIS GitHub repository;
- ✓ **Design of the dedicated tab in RDBFIS webapp** to upload the input for running the functions and visualise the output
- ✓ **Integration of the R package in RDBFIS**

TASK 4: Fine-tuning and further development of RDBFIS



- The package will incorporate information from bilateral meetings conducted during the implementation of **RDBFIS I and RDBFIS II**, along with insights from the **statistical workshop** for raising procedures. The package will be integrated into the RDBFIS platform
- **R package**
 - ✓ Raising of LFDs (landings, discards)
 - ✓ Estimation of age-length key
 - ✓ Estimation of discards ratio
 - ✓ Estimation of maturity (by age and length)
 - ✓ Estimation of sex ratio (by age and length)

❖ **Meeting for stat4RDBFIS:** Held on **7 July 2025** to discuss relevant topics related to the creation of an open-source R library for integration in RDBFIS and organize the work among the development team.

Thank you for your attention



EASME/EMFF/2020/OP/0021
Specific Contract CINEA/EMFAF/2024/3.1/01/SC08
Establishment of the Regional Database for the
Mediterranean and Black Seas
(RDBFIS-III, 01/04/2025 – 21/05/2026)



Project Coordinator: Dr. Irida Maina (HCMR)

Scientific Coordinator: Stefanos Kavadas